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The fitness costs and benefits of antibiotic resistance in drug-free microenvironments encountered in the human body

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Summary

The relationship between bacterial drug resistance and growth fitness is a contentious topic, but some antibiotic resistance mutations clearly have a fitness cost in the laboratory. Whether these costs translate into deleterious effects in natural habitats is less certain however. Previously, fitness effects of resistance mutations were mostly characterized in nutrient-rich, fast-growth conditions, which bacteria rarely encounter in natural habitats. Carbon, phosphate, iron or oxygen limitations are conditions met by bacterial pathogens in various compartments of the human body. Here, we measured the fitness of four different *rpoB* mutations commonly found in rifampicin-resistant bacterial isolates. The fitness properties and the emergence of these and other alleles were studied in *Escherichia coli* populations growing under nutrient excess and in four different nutrient-limited states. Consistent with previous findings, all four mutations exhibited deleterious fitness effects under nutrient-rich conditions. In stark contrast, we found positive or neutral fitness effects under nutrient-limited conditions. Two particular *rpoB* alleles had a remarkable fitness increase under phosphate limitation and these alleles arose to high frequencies specifically under phosphate limitation. These findings suggest that it is not meaningful to draw general conclusions on fitness costs without considering bacterial microenvironments in humans and other animals.

Introduction

The persistence of antibiotic-resistant bacterial pathogens in nature is a major problem. Antibiotic resistance is believed to result in a reduced fitness compared with drug-sensitive strains (Reynolds, 2000; Enne *et al.*, 2004; Gagneux *et al.*, 2006; Melnyk *et al.*, 2015). However, we know little about the fitness changes due to drug resistance in environments that can be linked to natural habitats of bacteria, which is crucial for our understanding of the persistence of antibiotic-resistant bacteria in the wild (Martinez *et al.*, 2007).

Unfortunately, the current gold standard method for estimating the fitness cost linked to drug resistance mutations (Melnyk *et al.*, 2015) involves a very unnatural situation. It entails head-to-head competition between susceptible and drug-resistant bacteria in antibiotic-free, nutrient-rich aerobic batch culture. However, like any biological phenotype, the fitness of an organism is dependent on many environmental factors such as nutrient availability. Effects of mutations can often be highly pleiotropic and their fitness contribution depends entirely on growth environments (Trindade *et al.*, 2012; Maharjan *et al.*, 2013). It is therefore very hard to disentangle the exact environmental conditions at which drug-resistant mutants exhibit fitness cost in batch cultures. A key question is whether the fitness measured using the so-called gold standard can be extrapolated to the real world, including different compartments of the human body and natural environments (Andersson, 2015; Durao *et al.*, 2016). Growth of most bacterial pathogens in the host is often limited by the availability of one or more essential nutrients such as phosphate, iron, oxygen or carbon (Matin *et al.*, 1989).

To investigate the environmental impacts on fitness associated with drug resistance, we use rifampicin-resistant (Rif^R) *Escherichia coli* isolates in controlled environments as a model system. Rifampicin is a clinically used antibiotic targeting the β subunit of bacterial RNA polymerase (RpoB) and blocks the path of elongating RNA (Campbell *et al.*, 2001). The target is the same across many species and even more remarkably, the same three conserved codons in RpoB are altered in Rif^R mutants in different bacteria (Goldstein, 2014). For example, changes at Ser531 gives resistance in *Mycobacterium tuberculosis*, *Staphylococcus aureus*, *Salmonella typhimurium* and *E. coli* (Goldstein, 2014). It has

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been speculated that these sites are common targets because they change RNA polymerase without causing major deleterious effects. Many other residues in RpoB can also be altered to give resistance, and all these changes are extremely pleiotropic and affect many phenotypes including fitness (Jin and Gross, 1989; Trindade *et al*., 2012).

The chemostat approach we use allowed us to fix four different ecologically important nutritional states (carbon, oxygen, inorganic phosphate and iron) at comparable growth rates. The limitation of growth rates by carbon source availability is a common feature of natural habitats (Ferenci, 2001). Likewise, oxygen limitation and the aerobic/anaerobic transition are central to the lifestyle of *E. coli* (Sawers, 1999). Phosphate limitation is also important, for example in lungs, and can trigger bacterial pathogenesis, in which mutations are involved [e.g., cystic fibrosis (Oliver *et al*., 2000; Rifat *et al*., 2009)]. The limitation for iron is likewise important in hosts that actively deplete iron, as in human body fluids (Kortman *et al*., 2014).

Here, we made two interesting observations that could explain the evolution and persistence of Rif^R mutants in nature. First, in stark contrast to previous findings, we find that Rif^R mutants generally have higher fitness under all the nutrient-limited conditions. In contrast, but consistent with previous studies, they are less fit in nutrient-rich environments. Second, we find that there is a consistency between fitness and allelic distribution of Rif^R in different environments, explaining the prevalence of particular mutant alleles.

Results and discussion

The fitness costs of Rif^R mutations are strongly dependent on the growth environment

To determine the fitness cost associated with Rif^R in different environments, we chose four different Rif^R mutants (Table 1) based on their prevalence in both clinical and experimental settings. Two of the isolates (MC4100RifB and MC4100RifC) were chosen because they are highly prevalent in clinical isolates of different bacterial species including *E. coli* and *M. tuberculosis* (Goldstein, 2014) and contained mutations affecting amino acid residues 526 and 531 of RpoB. Another strain (MC4100RifA) had a short deletion (Δ511–13), included in this study because deletions around this region are often detected in natural isolates. A fourth strain (MC4100RifD) with an I572L substitution was studied because it arose as a beneficial mutation without direct selection of Rif^R (Jin and Gross, 1989; Tenaillon *et al*., 2012). Table 1 shows the changes in *rpoB* and the growth rate changes in batch culture resulting from the mutations. All the changes gave a high MIC with rifampicin.

We investigated fitness (measured as the selection coefficient, *S*) of these strains under two different antibiotic free nutrient-rich environments by direct head-to-head competition with the wild-type in batch culture using a modified approach described in Lenski *et al.* (1991). Media for the two nutrient-rich environments included Luria-Bertani (LB) broth and minimal medium A (MMA) supplemented with 0.2% (w/v) glucose (Miller, 1972). Consistent with previous studies (Reynolds, 2000; Conrad *et al*., 2010), the fitness of all four drug-resistant strains in the nutrient-rich conditions was either unchanged or decreased for three mutants (Fig. 1A). We found a significant effect of mutations in glucose-rich minimal media for all four mutants (one-way ANOVA $F_{3,34} = 90.25$ and $P < 0.0001$). We found the ΔLSQ(511–13) mutant had the lowest deleterious effects, while the I572L mutant showed the highest fitness cost in both nutrient-rich conditions (Fig. 1A).

To investigate the impact of nutrient limitation on the fitness cost of Rif^R mutations, the selection coefficient of the same four Rif^R *E. coli* strains was determined under four different nutrient-limited environments using the same approach as previously described in Maharjan *et al.* (2006). We use phosphate (P), carbon (C), oxygen (O) or iron (Fe) limitation for this purpose. Limitation to

Table 1. Characteristics of four rifampicin-resistant *E. coli* strains.

E. coli strain	Codon change	Nucleotide change	Amino acid change	Region	Growth rate relative to wild-type: LB	Growth rate relative to wild-type: MMA	MIC (μg/ml)
MC4100	Wild-type	n.a	n.a.	n.a.	1	1	<10
MC4100RifA	CTG TCT CAG	Δ9bp@1530	ΔLSQ(511–13)	I	0.86 ± 0.01	0.98 ± 0.02	>1000
MC4100RifB	**C**AC>TAC	C1576T	H526Y	I	0.81 ± 0.01	0.94 ± 0.03	>1000
MC4100RifC	T**C**C>TTC	C1592T	S531F	I	0.80 ± 0.01	0.88 ± 0.03	>1000
MC4100RifD	**A**TC>CTC	A1714C	I572L	II	0.83 ± 0.05	0.85 ± 0.02	>1000

Regions or locations of mutations are based on the wild-type *rpoB* sequence of *E. coli* BW2952 (Ferenci *et al*., 2009). LB stands for Luria-Bertani broth and MMA for minimal medium A (Miller, 1972). We used glucose (0.2% wt/vol) as the carbon source in MMA. The minimal inhibitory concentrations (MIC) of rifampicin for four mutants were determined by growing in the LB liquid medium containing concentrations of rifampicin ranging from10 to 1000 μg/ml (10, 20, 50, 100, 200, 300, 400, 500, 600, 700, 800, 900 and 1000) in 96-well micro-plates at 37°C for 24 h.

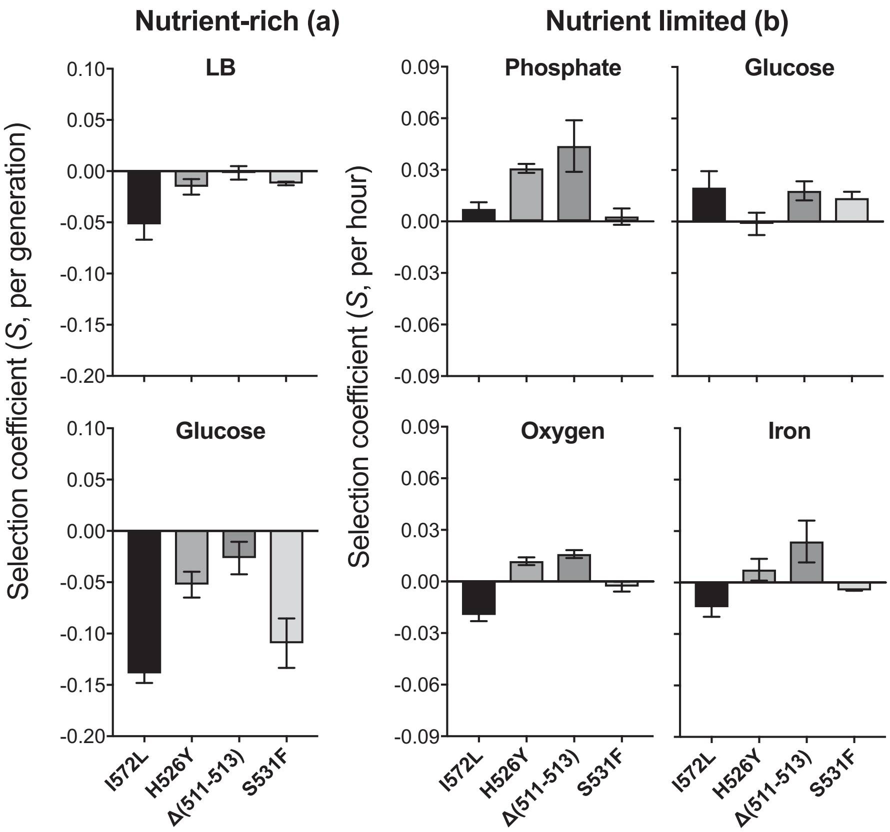

Fig. 1. Fitness measurements of four commonly found rifampicin-resistant mutants in nutrient-rich and nutrient-limited environments.
The fitness (selection coefficient) of four Rif^R mutants (MC4100RifA-D) with *rpoB* alleles ΔLSQ(511–13), H526Y, S531F or I572L derived from wild-type *E. coli* (Table 1) was determined in LB-rich medium and glucose-rich MMA medium in batch cultures (A), and four nutrient-limited environments (i.e., phosphate, carbon, oxygen and iron) in chemostats (B) by direct head-to-head competitions with the wild-type *E. coli*. For fitness assay in nutrient-limited environments, at least three chemostat competition experiments for each Rif^R strain were conducted as described previously in Maharjan *et al.* (2006). Briefly, each Rif^R strain and the wild-type strain were grown independently under the same nutrient-limited environments used for their isolation. Competitions were initiated by mixing 1:1 ratio of 15-h-old independent chemostat cultures (on the basis of OD_{600}) growing at a dilution rate of 0.1 h^{-1}. Dilutions of the mixed cultures were plated onto L-agar (for a total count of both strains) and LB-agar containing rifampicin (100 μg/ml, for Rif^R mutants counts). Samples were taken at time intervals of 1, 3, 6, 24 and 48 h. The reported fitness as selection coefficients (S) were based on the equation of Dykhuizen and Hartl (1983), in terms of Malthusian parameters determined from the linear slope of regression ln $[p(t)/q(t)]$, where $p(t)$ and $q(t)$ represent relative frequency of competing strains in the population at a given time point. For estimation of fitness in batch cultures, both mutants and wild-type *E. coli* were grown for 15 h in 5 ml LB in 25 ml tubes under identical conditions (i.e., 7°C, 200 r.p.m. orbital shaking). After mixing the equal amount of cultures (mutants and wild-type) based on their OD_{600}, 5 μl of mixed cultures were transferred into a 25 ml tube containing 5 ml fresh LB (1000-fold dilution) and allowed to propagate for 24 h. Initial and final frequencies of competing strains were then estimated by spreading as in chemostat competition experiments. The selection coefficient was then estimated by using the equation of Dykhuizen and Hartl (1983), $S=[\ln(n1f/n1i)-\ln\ (n2f/n2i)]/N$, where $n1_f$ and $n1_i$ are the number of the Rif^R mutants at the end and the beginning of the competition, $n2_f$ and $n2_i$ are the number of cells of the wild-type at the end and the beginning of the competition, and N is number of generation. We estimated the number of generations to be 15 based on the initial and the final cell number in the culture. Error bars represent the standard deviations from the mean fitness value calculated from at least three independent biological replicates in each condition.

these nutrients was achieved by limiting the amount of phosphate salt (K_2PO_4), glucose, aeration and iron separately in the chemostat with media as described in Wang *et al.* (2010) and Schliep *et al.* (2012). Each steady-state nutrient limitation, fixed by controlling the dilution rate, resulted in a specific growth rate of 0.1 h^{-1}. This rate, equivalent to a doubling time of 6.9 h, is 15- and 7-fold lower than the growth rate in LB or under glucose-excess conditions respectively.

As shown in Fig. 1B, none of the four strains exhibited loss of fitness. Instead, the Rif^R mutants were generally fitter than wild-type. It is worth noting that the influence of Rif^R was not uniform in the different conditions. In P limitation, all four mutants exhibited higher fitness but the extent of fitness changes was allele-specific. The S531F mutant showed the lowest fitness gain ($S = 0.003 \pm 0.005$), while the ΔLSQ(511–13) mutant showed the highest ($S = 0.044 \pm 0.015$). The difference in fitness was significant (one-way ANOVA, $F_{3,10} = 22.03$ and $P = 0.0001$). However, only two [H526Y and ΔLSQ(511–13)] mutants showed increased fitness compared with that of wild-type in both O and Fe limitation (Fig. 1B). The greatest fitness gain was also exhibited by these two strains in P limitation. These results suggest that Rif^R mutations are commonly beneficial under nutrient-limited environments, most likely contributing to their persistence.

Allele frequencies in different environments

To investigate whether the fitness differences in the five environments have a measurable impact on the persistence of the various alleles, we genotyped the Rif^R mutants present in these conditions. A total of 312 Rif^R isolates were sequenced (Fig. 2 and Supporting Information Table S1). For nutrient-limited cultures, isolates were collected from four-to-six 72-h-old replicate cultures growing in chemostats at the dilution rate of 0.1 h^{-1}. For glucose-rich cultures, we isolated Rif^R strains from replicate cultures growing exponentially in batch culture.

We detected 44 different mutations conferred a Rif^R phenotype in the 312 isolates (Supporting Information Table S1). The number of different *rpoB* alleles was found to be lower with P limitation than with the other three nutrient-limited environments (ANOVA, $F_{3,18} = 10.08$ and $P = 0.0004$). The allelic diversity of 0.35 (95% CI 0.26–0.43, $n = 6$) was the lowest in P limitation while Fe limitation showed the highest allelic diversity 0.77 (95% CI 0.52–1.02, $n = 4$). These results suggest P limitation may provide higher selective advantages for certain alleles of *rpoB*. In fact, we found that the Rif^R strain with H526Y as a result of C to T mutation at 1576 was dominant (54%) within the Rif^R population in P limitation. The same allele also occurred within Rif^R isolates in the other nutrient-limited environments but not as frequently as in P limitation. The frequency of Rif^R with H526Y ranged from 7% in Fe limitation to 20% in O_2 limitation (Fig. 2 and Supporting Information Table S1). The high proportion of H526Y mutant in P and other nutrient limitations is consistent with its higher competitive fitness (Fig. 1B).

Our results indicate that environmental settings strongly affect whether drug resistance is a cost or a benefit in the absence of selection pressure of antibiotics. We also demonstrated that fitness differences in the absence of antibiotics determine the emergence and maintenance of Rif^R mutations. We find that environments relevant to clinical settings have a major impact on the cost of rifampicin resistance. Phosphate limitation has the greatest positive impact, and this may be significant if a similar P limitation effect occurs for lung-affecting pathogens like *Pseudomonas aeruginosa* or Mycobacteria. There has also been previously demonstrated in the unexpected presence of *rpoB* mutations in selecting for fitness at high temperatures (Jin and Gross, 1989; Tenaillon *et al*., 2012; Rodriguez-Verdugo *et al*., 2013); these mutations results in fitness reduction at low temperature and fitness benefits at 42°C (Rodriguez-Verdugo *et al*., 2013). In general, the local environment may be a significant factor in setting the scene for resistance adaptation.

Although our results clearly demonstrate the remarkable environment-dependence of fitness effects associated with drug resistance mutations, the question remains to be answered at the molecular level – why are Rif^R mutants generally fitter under nutrient-limited environments? RpoB, a core component of RNA polymerase, has differentiated roles in environmental interactions and phenotypes with both nutritional and physicochemical states. This makes the precise contribution of Rif^R mutations particularly difficult to assess in complex host environments. The pleiotropic nature of RpoB probably accentuates the environmental differences. Previously, a study has pointed that there is a reduced demand for RNA polymerase in poor environmental conditions (Hall *et al*., 2011). This could be the reason why Rif^R mutants are able to ameliorate the deleterious effects of mutation under poor environments. Similarly, a study by Conrad *et al*. suggested that RNA polymerase mutants adaptively rewire the transcription network under certain growth conditions (Conrad *et al*., 2010). In addition, several other antibiotic targets (ribosomes, DNA gyrases, efflux systems, etc.) involve macromolecules that have multiple roles in cells, so the fitness variability may apply to these as well. Our results obtained in these settings (i.e., *rpoB* mutations and their fitness effects in different environments) cannot be extrapolated to other antibiotic resistance-conferring mutations. Further studies are needed to understand the adaptive role of RpoB under nutrient limitations.

Conclusions

A large number of studies (Enne *et al*., 2004; Gagneux *et al*., 2006; Melnyk *et al*., 2015) have investigated the fitness cost of rifampicin resistance in bacterial by using so-called the gold standard method (Melnyk *et al*., 2015) and the consensus conclusion was that Rif^R mutations are deleterious to bacteria in the absence of rifampicin. Thus, it is generally believed that sensitive bacteria will competitively eliminate the resistant counterparts in rifampicin-free environments unless they acquire compensatory mutations to ameliorate the costs (Comas *et al*., 2011). Our results suggest the opposite. We conclude that the fitness effects of Rif^R mutations are very sensitive to environmental conditions and are mostly

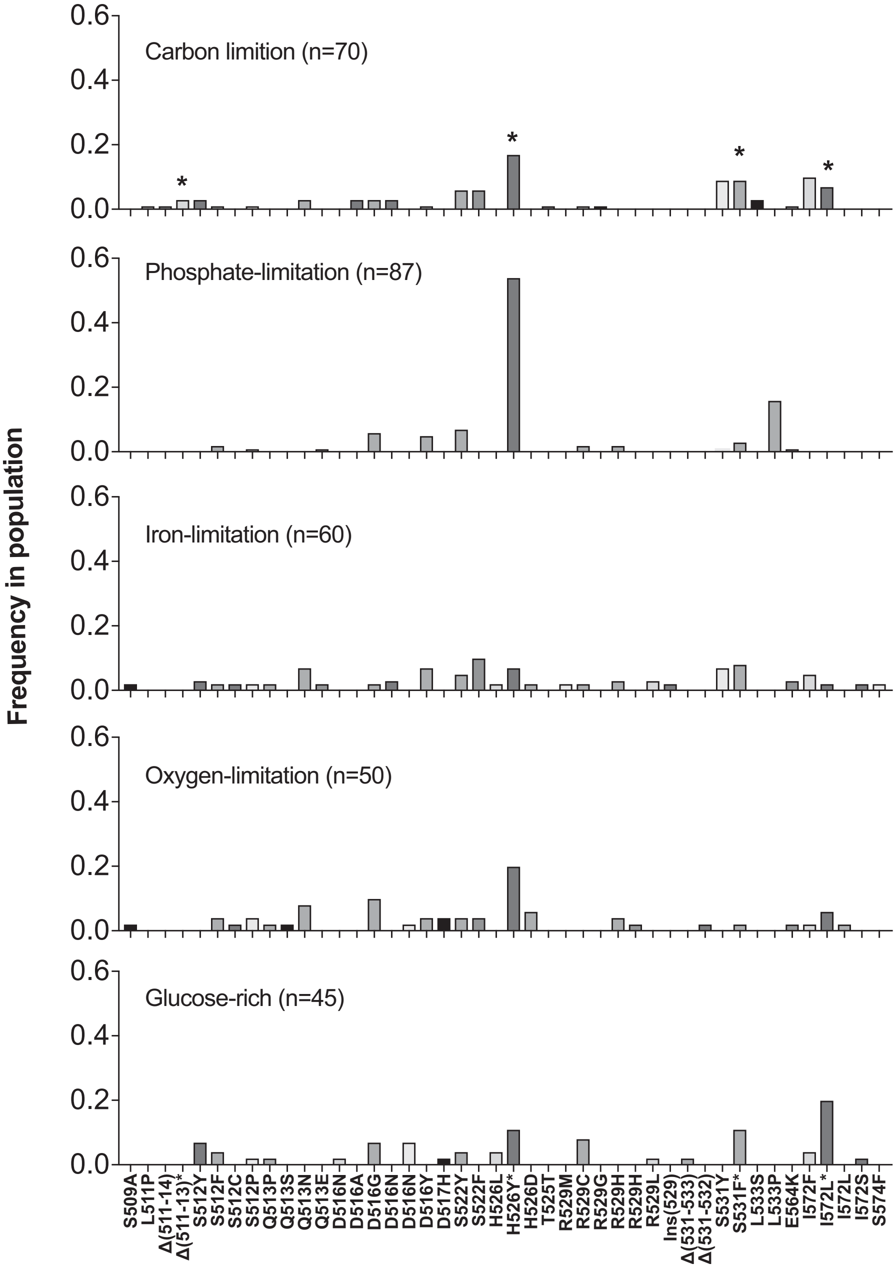

Fig. 2. Distribution of 44 different *rpoB* alleles conferring rifampicin resistance in *E. coli* in five different environments.
The allelic distributions are based on sequences of *rpoB* from a total of 312 rifampicin-resistant *E. coli* isolates randomly collected from five different environments, 70 from carbon-limited cultures (16, 9, 10, 12, 13 and 10 isolates from each of 6 populations), 87 from phosphate-limited (15, 14, 14, 13, 16 and 15 isolates from each population respectively), 60 from iron-limited (16, 15, 17 and 12 isolates from population respectively), 50 from oxygen-limited (16, 12, 12 and 10 isolates from each population respectively) limitations and 45 from glucose-rich or the unlimitation condition (12, 12, 9 and 12 isolates from each population respectively). These isolates were obtained from 72-h-old chemostat cultures growing at steady-state rate (0.1 h^{-1}) under the given nutrient-limited environments and from exponential growth phase bacterial cultures for the nutrient-rich condition. Allelic frequencies of mutations in *rpoB* conferring the Rif^R phenotype in *E. coli* in different environments were determined by dividing the total number of *rpoB* alleles observed by the total number of mutants found in each condition. The *rpoB* alleles were determined by sequencing PCR products of two regions of the *rpoB* in Rif^R mutants. Two sets of primers were used for both the PCR and sequencing. They are RpoBF1 (5′-gacagatgggtcgacttgtcagcg-3′) and RpoBR1 (5′-aggtggtcgatatcatcgactt-3′) and RpoBF2 (5′-tcgaaggttccggtatcctgagc-3′) and RpoBR2 (5′-ggatacatctcgtcttcgttaac-3′). The PCR products were purified using the Promega Wizard® PCR clean up kit (Promega, Sydney, Australia) and sequenced by a provider (Macrogen, South Korea) in both directions separately by using the forward and reverse primers. The sequence data were analysed with 4Peaks. *Rif^R strains harbouring *rpoB* alleles [ΔLSQ(511–13), H526Y, S531F and I572L) were used for fitness assays. All mutations, as well as their allelic frequencies in each condition, are listed in Supporting Information Table S1.

beneficial or neutral for bacteria under environmental limitations likely to be found in natural habitats. Rif^R alleles with increased fitness under nutrient limitation were also found to be dominant amongst the resistant bacteria both in experimental and clinical populations. A further conclusion is that in judging the cost of antibiotic resistance, the "gold standard" method of assessing fitness is highly misleading. This is perhaps not surprising, given that maximum growth rate in rich medium is an unnatural state in nature. Our results indicate that it is important to consider the impact of environments at the site of infection when modelling whether resistant mutants are likely to persist in hosts.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's web-site:

Table S1. Distribution of 44 different *rpoB* alleles conferring rifampicin resistance in *E. coli* in five different environments. The allelic distributions are based on sequences of *rpoB* from a total of 312 rifampicin resistant *E. coli* isolates randomly collected from five different environments, carbon (n=70), phosphate (n=87), iron (n=60) and oxygen (n=50) limitations and glucose-rich or nutrient un-limitation (n=45) condition. These isolates were obtained from four-to-six 72-h-old replicates populations (6 each carbon and phosphate cultures, 4 each in oxygen, iron and batch-unlimited cultures) growing at steady state rate under the nutrient-limited environments and from exponential growth phase bacterial culture for the nutrient-rich condition. Red type entries are mutated nucleotides. The *rpoB* gene was PCR amplified using the sets of primers, RpoBF1 (5′-tgggtcgacttgtcagcg-3′) and RpoBR1 (5′-aggtggtcgatatcatcgactt-3′), and RpoBF2 (5′-tcgaaggttccggtatcctgagc-3) and RpoBR2 (5′-ggatacatctcgtcttcgttaac-3′). The PCR products were purified and sequenced by a provider (Macrogen, South Korea) in both directions.